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Orange yield estimation using object tracking and 3D reconstruction

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ABSTRACT

The labor-intensive nature of agriculture, particularly in tasks such as yield estimation of fruits, is a significant challenge. Yield estimation is crucial for the better management of the resources and for taking adequate measures for the transportation, storage, and export of the fruits. It also helps the farmers to estimate the total pricing of the yield. However, counting fruits directly on trees for yield estimation presents an obstacle due to their dispersed nature and often dense foliage. Therefore, we propose that reasonably accurate fruit yield estimation can be automated with a handheld camera. The dataset is curated by capturing and annotating 1451 images of orange trees. The dataset is augmented and processed in different ways to evaluate the performance of YOLOv8 for the detection of oranges. Then the Byte tracker is deployed to track oranges in consecutive video frames. Further, we have classified the fruits into two categories, ripe and unripe using MobileViT. The 2D fruits detected by YOLOv8 are projected to a 3D space for a more detailed analysis of the scene. Subsequently, the clustering algorithm is applied to the 3D projections of the detected objects to estimate per tree yield. On images, YOLOv8 nano has achieved a precision of 78.2% and recall of 69.7% on the test set. Moreover, for ripeness stage classification, MobileViT has achieved an accuracy of 97.8% and 86.7% on a test set containing 2 classes and 3 classes, respectively. Testing our proposed solution on videos shows that the algorithm is achieving good results on trees with less leaf occlusion. This paper demonstrates that preprocessing techniques can aid the detection model to achieve high detection rates. Furthermore, per tree yield of an orange orchard can be estimated by using video input. This offers an automated solution to the laborious task of fruit yield estimation in agricultural settings, that can help to optimize orange production.

1. Introduction

The global shift towards fresh and convenient food consumption has significantly increased the demand for fresh fruits [23]. Citrus is one of the most widely grown fruit groups in the world. Oranges account for more than half of the world's citrus production. It is also the most widely traded citrus [7]. In 2021, citrus fruit worth of USD 26.41 billion were imported all over the world [6].

Top orange-producing countries are Brazil, China, and European Union with orange production of 15.3 million tons, 7.63 million tons, and 5.48 million tons respectively [27]. Pakistan, ranked as one of the largest citrus-producing nations, secures the 13th position in global cit-

rus fruit production [2]. In Pakistan, most farmers oversee small holdings, typically under 2 hectares. Punjab stands out with an average citrus farm size of 12.3 hectares, substantial compared to other crops. Citrus orchard sizes vary widely, ranging from under 1 hectare to 65 hectares in different regions. While a few large citrus growers exist, the landscape is mostly shaped by small and medium-sized growers [26].

In the fiscal year 2014–2015, total citrus fruit exports reached 393 thousand tonnes, valued at \$204 million, comprising around 16.4% of Pakistan's total citrus production. This marked a noteworthy threefold increase from 2000–2001, with 2009–2010 contributing \$99.4 million in foreign revenue.

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Despite being one of the largest citrus fruit producers, Pakistan's share of the global market remained less than 1% in 2008 [21]. A relatively modest proportion of citrus fruit (8–12%) is only directed towards international markets [26]. Not only Pakistan, but the citrus export of other countries also suffers due to poor economic strategies. For instance, in Nepal citrus is grown as the most important fruit crop, but due to the absence of proper cultivation and negligence has severely declined its production in recent years [9]. According to Bakari [4], new reforms are needed to refine trade and investment strategies in citrus trade of Tunisia.

This trend prompts an investigation into the dynamics of the citrus supply chain problems which entail a critical need for technological advancements to enhance the efficiency of the supply chain process. A pivotal stage in the journey from farm to market involves the manual counting and growth stage monitoring of citrus fruits, a process notorious for causing significant delays, fostering issues, and illicit methods for inaccurate estimates. This not only hampers the supply chain but also introduces opacity in citrus fruit transactions. To address these challenges, the implementation of an AI-based system for citrus fruit counting, growth grading, and shelf time estimation emerges as a strategic solution. Such technology not only replaces cumbersome manual methods but also mitigates associated problems, fostering transparency and efficiency throughout the citrus supply chain.

Previous research shows that fruit detection models are prone to environmental factors such as lightning, which changes throughout the day due to the positioning of sun in the sky. The performance of object detection models is also affected by the ripening stage of the fruits. Moreover, small-scaled objects are ignored by the detection models while downscaling the size of the input image. Some fruits in their initial stages have color similar to the leaves which can also affect the performance of the object detection models. What techniques for augmentation and preprocessing of the dataset can help object detection model to give good results in dealing with multi-scaled objects in varying lightning conditions. For video input, each frame is independently injected into the detection model. This leads to the recounting of the single object in multiple frames. Therefore, there is need for the implementation of the object tracking model to keep track of the objects to avoid recounting.

Detection model can also be used for the ripeness classification of the oranges. Using a single model for both the tasks can reduce the inference time which will be taken by a using a separate classifier. However, can a detection model give as good results as a sophisticated classification model. Calculating the overall yield doesn't give the in depth detail of the orchard. Therefore there should be an algorithm that can give per tree yield of the orchard which can further be used for efficient use of resources such as water and fertilizers.

The main contributions of this research are as follows:

- **Dataset Collection and Annotation:** Around 1400 images of orange trees are taken from the orchards of National Agriculture Research Center (NARC) and Guangxi Normal university. These images are annotated in detail where each orange is annotated as a separate instance. Dataset is also annotated for fruit ripeness stage classification into 3 labels.
- **Evaluating the performance of object detection model using efficient preprocessing:** Images are augmented by varying the brightness and saturation to mimic lighting changes occurring due to the sun. The images are tiled in a 2x2 grid and resized to 640x640 to preserve the features. Variants of YOLOv8 are trained and evaluated on the dataset that is processed in 4 different ways.
- **Implementing a dictionary tracking algorithm:** Using Byte tracker and dictionary management to track each fruit through each tile of the image and through consecutive frames of the video
- **Evaluating the performance of vision transformer classifier:** YOLOv8 and MobileVit is trained and evaluated to classify fruits into different ripeness categories.

- **Proposing 3D clustering for per-tree yield of oranges:** Detected fruits are projected to a 3D plane for the estimation of the placement of the oranges in real world. Then by using K-means clustering on the 3D point cloud, oranges are assigned to a tree to estimate per-tree yield.

The rest of the paper is organized as follows: The related work is discussed in Section 2, a description of the dataset is given in Section 3, the methodology is presented in Section 4, Section 5 contains the results and discussion, and Section 6 presents the limitation and future work.

2. Related work

Computer vision has provided efficient solutions to revolutionize these agricultural practices. In past years, research has been done to automate the task for yield estimation. For instance, Asakereh et al. [3] has compared the performance of YOLOv2, YOLOv3, and YOLOv4 for detection of oranges. YOLOv4 has outperformed YOLOv2 and YOLOv3 with an F1 score of 92%. A segmentation technique has also been proposed by Liu et al. [15]. They have used the DLNet double-layer occlusion segmentation model to segment green fruits in an obscured environment.

In Liu et al. [18], enhanced object detection in citrus orchards with a model combining coordinated attention and cross-scale connection with weighted feature fusion. Achieving 98.4% precision, the model surpasses YOLOv5 in accuracy, making it a significant advancement in precision agriculture for real-time, multiple fruit target detection. Lyu et al. [19], introduced YOLOv5-CS, a lightweight model, for green citrus detection and counting in orchards. Utilizing image rotation and convolutional block attention, the model achieves 98.23% detection accuracy and 28 FPS frame rate on an AI edge system, marking a step forward in efficient green citrus counting.

In Lin et al. [14], an improved YOLOv5s algorithm combined with DeepSort for citrus counting, tackling shading, and lighting challenges in orchard videos is proposed. This enhancement increases recognition accuracy by 3.51% and tracking accuracy to 90.83%, demonstrating significant improvements in real-time citrus detection and tracking. Gao et al. [8] developed a multi-class fruit detection method using Faster RCNN for apples in dense-foliage orchards. This method, distinguishing apples in various conditions, indicates a mean average precision of 0.879, aiding in effective robotic picking strategies and damage prevention.

Chakraborty et al., in 2023, developed an automatic machine for postharvest citrus operations, leveraging a custom lightweight CNN model, SortNet. This machine performs tasks like washing, sorting, and grading with high accuracy, showcasing a sustainable solution for postharvest automation in citrus fruits. Lastly, Gao et al., in 2022, developed a novel method for apple detection and counting in orchards, using the YOLOv4-tiny network and CSR-DCF algorithm. Achieving a mean average precision of 99.35% for fruit and trunk detection, this method provides reliable yield data for apples, representing a significant advancement in orchard environment monitoring.

The depth camera (Intel REalSense d435) is mounted on a remote-controlled vehicle for data acquisition [28]. YOLO v4 and trunk tracker are utilized for fruit detection and fruit tracking. For the elimination of the counted fruits for the previous row, the depth frame of the camera is used. If the distance between the detected object and the camera is more than 3 m, then the fruit belongs to the previous row. Moreover, for the removal of the fallen fruits, confidence scores are observed to be low. An average accuracy of 95% is achieved in the original video.

Abeyrathna et al. [1] has targeted apple orchards for object detection and tracking. Data is collected using a 3D camera mounted on a forward-moving vehicle. YOLOv4, YOLOv5, YOLOv7, and EfficinetDet are used to perform the task of fruit detection. Then RGB coordinates are aligned with the depth channel obtained by the 3D camera. For tracking of fruits across frames of videos DeepSort is implemented. The highest detection accuracy is obtained by YOLOv5 and YOLOv7.

YOLOv7 with a channel-based attention module and convolutional block attention module is proposed in Hu et al. [11] for apple detection, and cascaded sort with SURF detectors is used for more accurate and real-time apple tracking. Flying robotic system (FRS) captured 100 apple images for apple counting in Yildirim and Ulu [29]. MobileNet SSD and Faster RCNN are used for object detection. Faster RCNN has obtained an accuracy of 93%.

Two infrared cameras along with lidar sensors are developed on a moving vehicle for the collection of data [16]. lidar gives a more accurate depth channel. Multiple lidar scans have done point cloud registration. The scene is reconstructed in a 3-dimensional space and the camera pose is estimated by ORB-SLAM3 algorithm. For object detection in RGB, YOLOv5 is utilized. Further Information obtained from lidar and RGB is combined using fruit depth association and cluster filtering. The average localization error is 21.1 mm with 90% of the errors being less than 30 mm.

Qi et al. [22] shows that YOLOx tiny gave a detection accuracy of 92.71% with minimum inference time. A new architecture nanoFet-m-plus is proposed to classify the ripeness stage into 5 levels and deepsort is used for object tracking. YOLOv5 along with the contextual transformer self-attention module with SIOU loss is used by Lin et al. [14]. The introduction of the attention module has increased the performance by 3.51%.

Berry-shaped fruits have two types of annotation, i.e. MOTs style annotation (complete cluster is labeled as one unit) and COCO style annotation (each berry unit is labeled independently). Liu et al. [16] has used both annotations for the early extraction of grape phenotypic traits. PointTracks detects the grape clusters and YOLACT along with the spatial embeddings detects each berry unit. It has given a detection result of 93.85%.

In Riccardi et al. [24] high-precision point cloud data from the strawberry greenhouse is collected by a terrestrial laser scanner. It has used a novel 3D fruit descriptor that captures the topology of the fruit surroundings and a cost function that combines positional distance, descriptor distance, and radius variation. Further, Hungarian method is applied to find the optimal fruit associations. Zhang et al. [30] has used Siamese convolution autoencoder combined with DeepSort for object tracking.

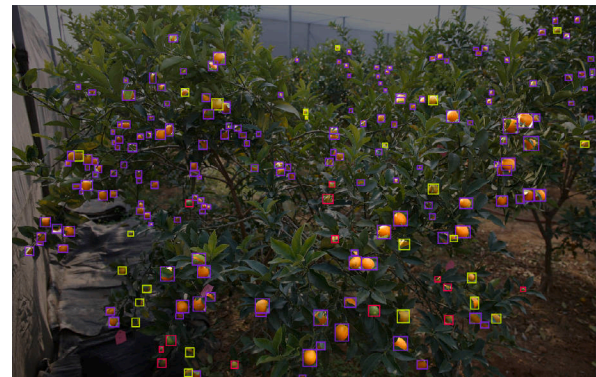
3. Dataset

3.1. Collection and annotation

Orange trees flower from late February to early April in most areas of Pakistan. The fruit set period starts during April and May. The harvest period for oranges occurs from December to March [13]. Therefore the duration of data collection was for a few months from August to December. In this way, we were able to collect images of oranges over their different ripeness stages.

The dataset is obtained from two different sources. The first source, comprises images captured with a Kinect V2 depth camera, providing enhanced RGB data. This dataset includes a variety of conditions, such as differing lighting and angles, to simulate real-world scenarios, and the images are captured from the orchards of NARC [5]. 694 images are taken from a far view, which can be seen in Fig. 1a.

We also requested Guangxi Normal University, a Chinese university, to share their dataset for object detection of oranges. The dataset that they shared had different annotations than ours so we re-annotated the images in their dataset so that we can use it for our research. The dataset comprises images taken from a closer view, which can be seen in Fig. 1b. The dataset consists of a total of 1451 images taken in different lighting and weather conditions. Further, from this dataset 80% images are used for training and 10% is used for testing and 10% is for validation purposes. These 1451 images are manually annotated in detail using Roboflow for bounding boxes and for ripeness stage classification. Oranges throughout their life cycle go through different stages of Ripeness. The color of the oranges changes throughout their life which helps to



(a) Images from NARC dataset



(b) Images from Chinese dataset

Fig. 1. Image samples from the dataset.

identify the stage of ripeness. Initially, the dataset was annotated in two stages ripe and unripe. But during the testing phase, we felt the need to introduce a new class, partially-ripe, as there is a stage in its life cycle when it is not fully ripped and can not also be classified as unripe. Therefore the classification dataset is annotated twice, one dataset consists of two classes, i.e. ripe and unripe and the other one consists of unripe, partially-ripe, and ripe.

For testing the algorithm on videos, 2 videos of the orange trees are taken in orchards in a linear motion. One video only consists of one tree and the other video consists of 3 trees in a row. These trees are highly loaded with oranges and branches are also densely covered with leaves. These two videos are taken from an orchard in Sargodha. However, a different video is shot at different angles and in nonlinear motion to evaluate the performance of the detection, tracking, and classification model.

3.2. Augmentation and preprocessing

Deep learning models require a large dataset to produce good results. The dataset consists of 1451 images, and each image consists of multiple orange instances. Before the augmentation of the dataset, each image is tiled in a 2x2 grid creating 4 images from a single image. After tiling, some images will not contain oranges in it. These images will also act like the negative samples.

To increase the dataset, several images are flipped and rotated at different angles, and parameters like sheerness, hue, saturation, brightness, and exposure are altered. Furthermore, the images were blurred and noise was introduced to make the model more robust. The images are tiled in a 2x2 grid to create 4 tiles of images. After augmentation, each image tile is resized to 640x640 pixels. By tiling and augmentation, the total number of 640x640 images reached 17,412 total images.

For the classification of oranges, a separate dataset is created by cropping oranges from 17,412 augmented images. Each image in this dataset

consists of only one orange. Oranges in the dataset are of different sizes, the ones in the background appear smaller in size whereas the ones close to the camera appear to be bigger. So cropping of these oranges will result in images of varying sizes. Therefore all the images are resized into 50x50 pixels to have a constant image size throughout the dataset. The dataset consists of a total of 143988 oranges consisting of 91277 unripe oranges and 52711 ripe oranges.

4. Methodology

The first step is to collect and annotate the data to detect the maximum number of fruits in each frame of the video. Each detected fruit is assigned a unique ID. Then, this ID will be maintained across consecutive frames of the video through an object tracking algorithm. Each object will be classified in its ripeness stage using a lightweight classification model. To further refine the object count by removing recounted objects and to estimate per tree yield, 3D coordinates of the fruits are calculated. Lastly, these 3D points are clustered to obtain tree counts and objects that belong to individual trees.

4.1. Object detection

The main aim of the object detection stage is to detect the maximum number of fruits in each video frame. For the detection of oranges in orchard we have used YOLOv8 [12]. Ultralytics introduced YOLOv8 in January 2023, enhancing the YOLO architecture with several key improvements. YOLOv8 features a C2f module that combines high-level features with contextual information, improving detection accuracy. It employs an anchor-free model with a decoupled head, allowing independent processing of objectness, classification, and regression tasks, thereby refining overall accuracy. The output layer uses a sigmoid function for objectness scores and a softmax function for class probabilities. YOLOv8 also introduces a semantic segmentation model, YOLOv8-Seg, which delivers state-of-the-art performance while maintaining speed and efficiency. The model utilizes advanced loss functions like DFL and CIOU for bounding-box loss and binary cross-entropy for classification, which boosts performance, especially for smaller objects.

Ultralytics offers five versions of YOLOv8—nano, small, medium, large, and extra-large—each optimized for different use cases and hardware requirements. As we aim to detect oranges in real time, medium, small, and nano versions of YOLOv8 are trained. As the objects that we aim to detect are very small, so tiling is performed on each image for better detection results. Original images in the dataset have a resolution of 5760x3840 or 6000x4000. For tiling, the image is split in a 2x2 grid resulting in 4 sub-images. These object detection models are trained in four different ways.

1. Tiling is not performed for both training and testing: All images are downscaled to 640x640 resolution for both training and testing.
2. Tiling is performed only during inference: For training, the complete image is downscaled to 640x640. Whereas while testing the image is tiled into 4 sub-images. Each sub-image is then resized to 640x640 pixels. The hypothesis behind this method is that if the model learns to perform well on low-resolution images it can give even better results on high-quality inputs
3. Tiling is performed only during training: Splitting the image into 4 images and running an object detection model on each sub-image will increase the computation cost 4 times. So tiling is only performed on the training set as a method of augmentation. For testing, the whole image is downscaled to 640x640.
4. Tiling is performed for both testing and training: For both training and testing, images are split into 4 sub-images, then each sub-image is further resized to 640x640.

The object detection model runs on each tile independently. Predictions from each sub-image can be mapped to the original image using



Fig. 2. Red zone is used to represent the area that is used to identify the lost objects.

the equations (1) and (2), where x_{org} , y_{org} are the coordinates of the bounding box in the original image, i be the split number of the image. The split number is assigned incrementally from top-left with $i = 0$ to bottom-right $i = gridsize^2$. As we are splitting our image into 4 sub-images, so the grid size is 2, and W and H are the height and width of the image before slicing.

$$x_{org} = x_i + \left[(i \% gridsize) * \left(\frac{W}{2} \right) \right] \quad (1)$$

$$y_{org} = y_i + \left[(i / gridsize) * \left(\frac{H}{2} \right) \right] \quad (2)$$

The object detection models are trained on an Intel i7-12700k CPU (3.61 GHz), 64 GB RAM, Nvidia 3090 GPU (24 GB), CUDA 12.0 for GPU acceleration. Models are trained with a learning rate of 0.01, batch size of 16, Adam optimizer, 0.937 momentum, and weight decay of 0.005 for 50 epochs.

4.2. Object tracking

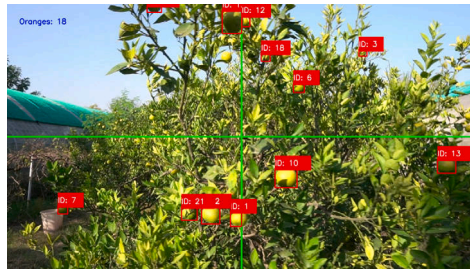
The Byte tracker of YOLOv8 runs independently on each tile of the image, tracking only the objects in individual tiles. It effectively tracks the object as long as it's in the same tile. However when an object jumps from one tile to another a new ID is assigned to the newly detected object. This problem has been tackled by keeping track of lost and newly detected objects. The region around the area from where the image splits is considered a danger zone as highlighted in Fig. 2. This region lies at 50 pixels to the splitting lines in each tile. Tracking of the objects that are lost or are first detected around this region is maintained to avoid the assignment of new IDs to these objects (Figs. 3, 4). Euclidean distance is calculated between these lost and newly detected objects to re-identify the lost objects as given in equation (3), where $d(L, N)$ represents the distance between the lost orange and the new detected orange, x_l and y_l be the x and y coordinates of the center of the lost orange, and x_n and y_n be the x and y coordinates of the center of the new detected orange. If an object can not be identified within 10 frames after it is lost, the object is dropped from the tracking list.

$$d(L, N) = \sqrt{(x_l - x_n)^2 + (y_l - y_n)^2} \quad (3)$$

4.3. Ripeness stage classification

For ripeness stage classification, the performance of the YOLOv8 classifier and a lightweight vision transformer MobileViT XXS is compared.

MobileViT [20] is a hybrid architecture that combines Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs) to achieve efficient and powerful performance on mobile and edge devices. It uses the locality and translation invariance of CNNs alongside the global receptive field of ViTs. MobileViT consists of a series of MobileViT blocks that are strategically interleaved with convolutional layers. Each block



(a) Orange ID:1 is detected in the bottom left quadrant



(b) Orange ID:1 is lost while shifting from the bottom left quadrant to the bottom right quadrant



(c) Orange ID:1 is re-identified in the bottom right quadrant

Fig. 3. Results of tracking algorithms.

(a) Detection results on original video frame



(b) Detection results on tiled video frame

Fig. 4. Qualitative results of Object detective model before and after tiling.

starts with a local feature extractor using depth-wise separable convolutions that significantly reduce the parameter count and computational cost. This is followed by a transformer-based module that processes the features in a sequence-to-sequence manner, capturing long-range dependencies. The output of this module is then integrated and processed through point-wise convolutions to combine the features effectively. This design allows MobileViT to maintain high accuracy while being compact and efficient, making it particularly suitable for deployment on devices with limited computational capabilities.

Each detected object is resized to 50x50 pixels to be used as input for the classifier. MobileViT is trained using stochastic gradient descent as an optimizer, with a learning rate of 0.01 and categorical cross-entropy as a loss. For inference on video, it will become costly to run the classifier on each object of every frame of the video. To reduce the computational cost, MobileViT runs depending on the following condition:

1. The object is detected for the very first time.
2. If the model was not very confident in the previous classification of the detected object (confidence < 0.8).

4.4. 3D reconstruction

For estimating per tree yield, 2-dimensional coordinates of fruits are not enough. In video input, coordinates of each fruit will change with a change in the angle and motion of the camera. To get an estimation

of the location of the fruits, they have to be projected on a 3D space. Therefore, detected fruits are projected to a 3D space by following the methodology in Liu et al. [17]. For this purpose, COLMAP, a structure from a motion algorithm, is used. As 3D reconstruction is a very computationally costly process, COLMAP constructs a 3D scene from a set of 2D images.

COLMAP [25] is a widely-used photogrammetry and Structure-from-Motion (SfM) software that excels in reconstructing 3D models from a series of images. It is an open-source tool that integrates both feature extraction and matching with camera pose estimation and dense 3D reconstruction. The software is highly versatile, supporting various camera models and formats, which makes it suitable for a wide range of applications, from academic research to commercial projects. By leveraging advanced algorithms like bundle adjustment and depth map fusion, COLMAP can produce highly accurate and detailed 3D models, even from datasets with challenging lighting and texture conditions. The software supports multi-GPU processing, which significantly speeds up the reconstruction process, making it ideal for handling extensive collections of images. Additionally, COLMAP provides options for sparse and dense reconstruction. Its robust feature matching and triangulation capabilities ensure that the resulting 3D models are both geometrically accurate and visually compelling.

SIFT (Scale-Invariant Feature Transform) is used to calculate and describe these detected features. These descriptors are crucial as they must be invariant to changes in scale, rotation, and lighting. After features

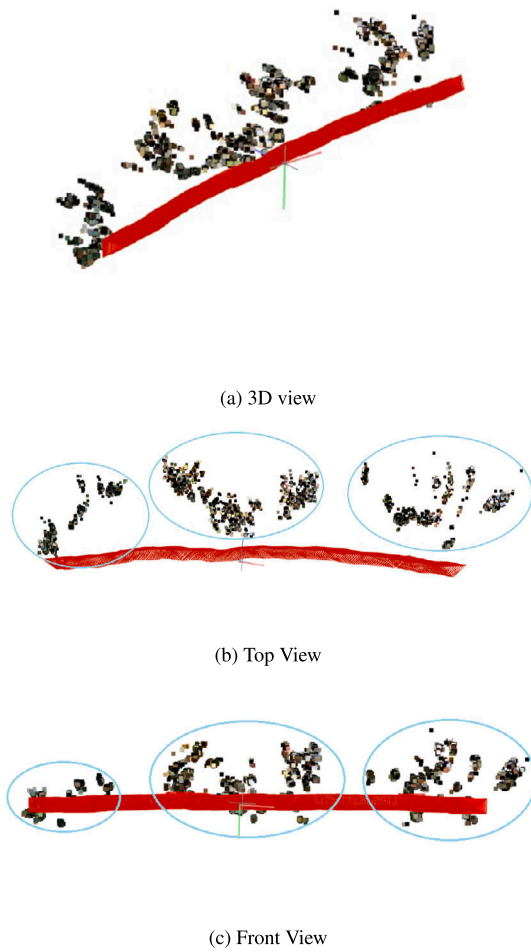


Fig. 5. Different views of 3D project of oranges.

have been extracted from each image, the next step is to find correspondences between features in different images. This is typically done using nearest neighbor search, where features from one image are matched with features from another based on their descriptors. With a set of matched features across multiple images, a sparse 3D point cloud is reconstructed. Then the camera parameters (like position and orientation) are estimated using structure from motion (SFM) and the sparse structure of the scene simultaneously.

COLMAP provides both a command-line interface and a graphical user interface, making it accessible for automated scripting and interactive modeling. Its flexibility and comprehensive toolset make it popular for projects reconstructing 3D models from photographs or video frames. The features are only calculated around the detected object by adding a margin of 5 pixels on each side of the object. By doing that, the computational cost has dropped drastically. To further reduce the computational expense only every third frame of the video is used for the stereo view 3D reconstruction. COLMAP calculates feature descriptors of the objects using SIFT. Further, it uses a plane sweep algorithm for a stereo-view 3D reconstruction (Fig. 5).

4.5. 3D points manipulation and object assignment

As 3D point annotations are not available, we can unsupervised algorithms to estimate which fruit belong to which trees. Trees planted in orchards have an approximately fixed distance between rows of trees and a fixed distance between the trees. Moreover number of trees in each row is also known. This is exploited to obtain the per-tree yield of the fruits. From the assumption that the fruits that belong to the same

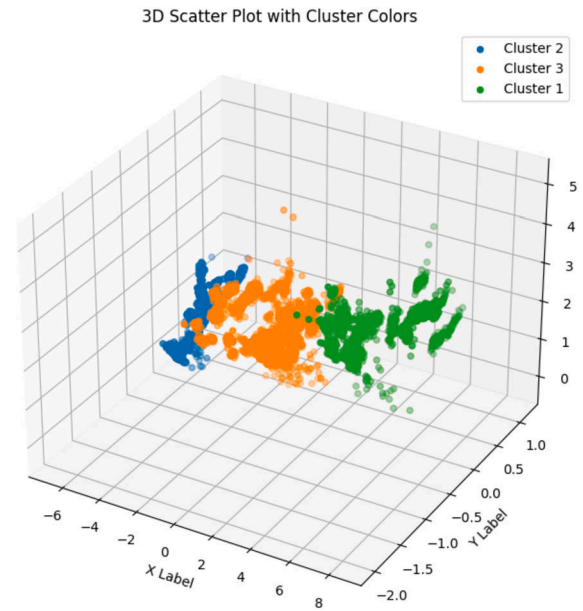


Fig. 6. Clustering of oranges in 3D plane to estimate per tree yield. Each cluster represents fruits that are estimated to belong to the same tree.

tree are closer to each other than the fruits from the other trees, the fruits that are closer to each other are clustered together.

For that K-means clustering is implemented on all the 3D points obtained by 3D reconstruction with k being the number of trees in a row. For instance, if a row consists of 10 trees then $k = 10$.

K-means clustering is an algorithm that partitions data into k clusters by assigning each point to the nearest centroid and then recalculating centroids as the mean of the points in each cluster. This process repeats until the centroids stabilize. It is simple and fast, but requires specifying the number of clusters in advance and can be sensitive to initial centroid placement.

Each cluster represents a tree. We assume that the oranges belonging to a cluster also belong to a single tree. After clustering the oranges in a 3-dimensional space, they have to be mapped to the 2D plane to identify the IDs of the fruit in the 2D video frames.

Mapping 3D points back to the 2D image can help us obtain the per-tree yield and to remove the recounted fruits that are missed by the object tracker. Therefore the 3D points are mapped to the 2D images. Each object contains multiple feature points and each feature point belongs to a cluster. So the cluster that is assigned to the object is the most occurring cluster in its feature points. As clusters can be in a random order, we can not assign the tree ID based on the cluster number. Therefore the mean of all the 3D points of the cluster is calculated to estimate the center of the tree. These centers are then sorted in ascending order based on their x values. Depending upon the x values of the center the tree ID is assigned. In this way, we can have an incremental ID for trees in a row (Fig. 6).

Some fruits do not have any feature points. Tree assignments of such fruits are done by K nearest neighbors [10]. Depending upon the tree ID of the five closest fruits, the unassigned fruit is assigned a tree ID.

If a feature point belongs to more than one fruit then it indicates that the fruit is recounted. If we use a single feature point for the detection of reappearing fruits, it can cause false removal of the object as the target fruits can have similar features. Therefore, those fruits that have the same feature points for more than half of the total feature points of the objects.

5. Results and discussions

Object detection models are evaluated on test and validation sets using precision, recall, mAP50, and mAP50-95. Similarly, the classifi-

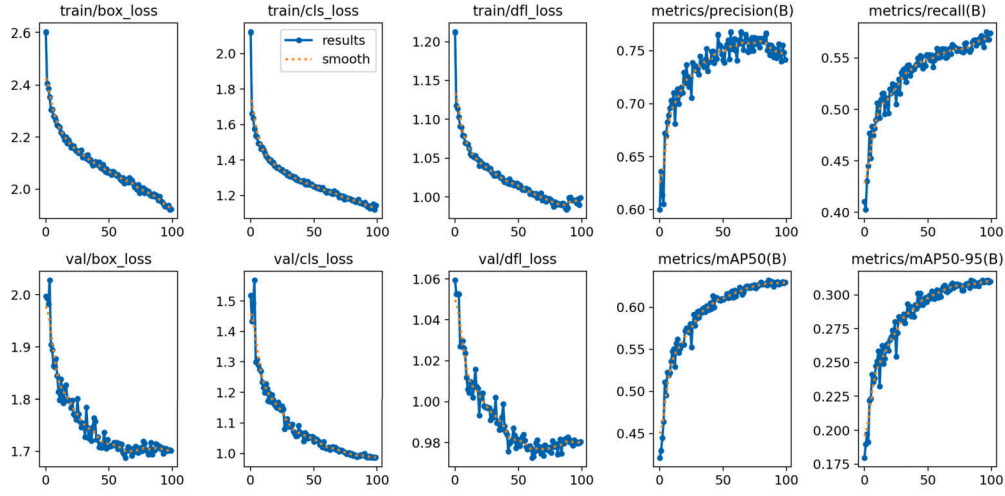


Fig. 7. Training curves of YOLOv8(nano) for original dataset.

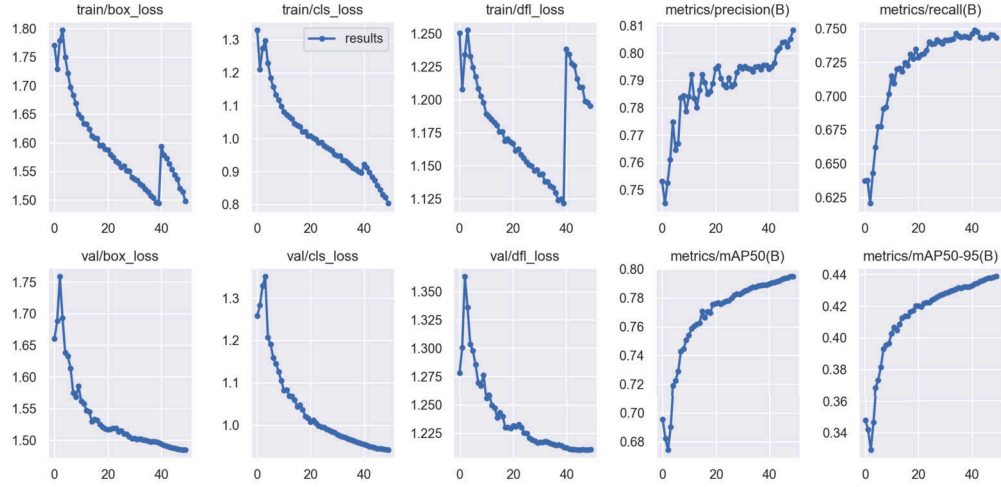


Fig. 8. Training curves of YOLOv8(nano) for tiled dataset.

cation models are evaluated on test and validation set using accuracy, precision, recall, and F1-score by using equations (4), (5), (6) and (7), respectively. In equations (5) and (6), TP is the number of true positive instances, FP is the number of false positive instances, and FN is the number of false positive instances.

$$Accuracy = \frac{No.ofCorrectPredictions}{TotalPredictions} \quad (4)$$

$$Precision = \frac{TP}{TP + FP} \quad (5)$$

$$Recall = \frac{TP}{TP + FN} \quad (6)$$

$$F1 - score = \frac{2 * Precision * Recall}{Precision + Recall} \quad (7)$$

Images are tiled to preserve the features of small-scale objects. Instead of resizing the complete image to 640x640 resolution, each image is tiled in a grid of 2x2. Each tile is then scaled to 640x640 resolution, in this way the small-scaled objects will not be lost while downsampling the image, resulting in high detection results. Using high-resolution images instead of tiling can also give similar results but it requires very high computational power to run the model. Tiling can reduce the computational cost at a given time but may increase the total time required to run the model on all tiles of the image. However, this problem can be solved by using running models on all tiles in parallel to minimize inference time.

YOLO variants are trained on the tiled and non-tiled datasets. They are evaluated on both the validation set and test set using multiple evaluation metrics. Medium, small, and nano variants of YOLOv8 that are trained and tested on non-tiled images show that YOLOv8 medium outperforms small and nano with the highest precision, recall, and mAP on the test set among the variants. When these three variants were tested on the tiled dataset they produced lower precision and higher recall. The nano version of YOLOv8 is trained on tiled images and tested on both tiled and non-tiled test and validation sets. When this model is tested on tiled images it has outperformed all the models with the highest precision, recall, mAP50, and mAP50-95 of 82.2%, 74.6%, 78.8%, and 46% respectively on the validation set. Similarly, it has also outperformed other models on the test set with a precision of 78.2%, recall of 69.7%, and mAP50 of 76%. However, when this model is tested on the non-tiled dataset, it has given the worst results among all the models.

It shows that the tiling at training helps in increasing the overall performance of the models if the testing is performed on the tiled images. Whereas when the same model is tested on non-tiled data, a decrease in the number of detected oranges is reported. But when non-tiled trained models run on tiled images, it increases the number of detected oranges but also increases false positives. Training curves of trained on non-tiled and tiled datasets can be seen in Fig. 7 and Fig. 8, respectively.

These models are further tested on videos obtained from two different orchards in two different cities of Pakistan, i.e. Islamabad and Sargodha. YOLOv8, trained on original data, is tested on video frames without tiling and it has given very poor detection results in terms of

Table 1
Detection results on the validation set.

YOLOv8 Variants	Training setting	Testing setting	Precision (%)	Recall (%)	mAP50 (%)	mAP50-95 (%)
Nano	Non tiled	Non tiled	74.7	56.6	63.2	31.3
Small	Non tiled	Non tiled	78.2	61.2	67.7	34
Medium	Non tiled	Non tiled	76.8	59.7	65.2	32.8
Nano	tiled	tiled	82.2	74.6	78.8	46
Nano	Non tiled	tiled	3.8	60.7	65.7	32.2
Small	Non tiled	tiled	72.1	65.5	68.2	34
Medium	Non tiled	tiled	72.1	65.5	68.2	34
Nano	tiled	Non tiled	71.4	47.1	53.8	22.4

Table 2
Detection results on the test set.

YOLOv8 Variants	Training setting	Testing setting	Precision (%)	Recall (%)	mAP50 (%)	mAP50-95 (%)
Nano	Non tiled	Non tiled	72.6	56.8	63.6	30.3
Small	Non tiled	Non tiled	75.1	61.1	67.5	33.4
Medium	Non tiled	Non tiled	76.9	59.1	66.3	32
Nano	tiled	tiled	78.2	69.7	76	38
Nano	Non tiled	tiled	68.8	57.6	61.7	25.9
Small	Non tiled	tiled	67.9	61.8	63.6	25.8
Medium	Non tiled	tiled	65.6	60.6	0.619	24.7
Nano	tiled	Non tiled	69.1	48.6	56	23.1

Table 3
Evaluation Results of Classification Models.

Model	Dataset	Validation Accuracy (%)	Test Accuracy (%)	Test Precision (%)	Test Recall (%)	Test F1-score (%)
YOLO v8 (nano)	2 classes	68.3	70.7	70.9	69.4	70.1
MobileVit XXS	2 classes	97.5	97.8	97.6	97.5	97.7
MobileVit XXS	3 classes	85.2	86.9	86.9	87.4	86.6

recall. It fails to detect oranges when the lightning conditions are different than those of the training set. Moreover, it also lacks the ability to detect small fruits (Tables 1, 2).

When this same model is tested on tiled frames, it has given high detection results as the model was trained to detect oranges from low-quality images. So when such a model is tested on high-quality frames it also detects small objects from the image. However, it has also given false positives by detecting similar color leaves as oranges.

YOLOv8, trained on tiled images, failed to perform well on non-tiled video frames, as the model is trained on high-quality images. Hence it has poor performance on low-quality data. But this model is giving high detection results on tiled images with minimum false positives.

For ripeness classification, YOLOv8 was able to achieve an accuracy of 70.7% whereas the mobileVit achieved an accuracy of 97.8% on the dataset consisting of 2 classes whereas it achieved an accuracy of 86.9% on the dataset consisting of 3 classes. MobileVit is robust to lighting changes which is observed in orchards throughout the day. Results are given in Table 3.

Confusion Matrices of MobileVit are given in Fig. 9a and Fig. 9b on the dataset with 2 classes and 3 classes respectively. From the confusion matrix, it can be seen that the precision and accuracy of MobileVit, trained on data with 3 labels, has significantly dropped due to the label being partially-ripe. This class is affecting the results of the model due to the issue of class imbalance. Whereas when we annotated the data with 2 labels, there was a significant increase in the accuracy and precision of the classification model.

For the testing on the video, the tracking algorithm has effectively tracked the oranges in the frame tiles of the video. The dictionary management algorithm helped to keep track of the oranges across the different tiles of the frames using Euclidean distance. This helped in remapping the objects across the tiles of the image. This algorithm performs well when the object is moving at a small pace but tends to skip a

Table 4
Results of Orange Counting.

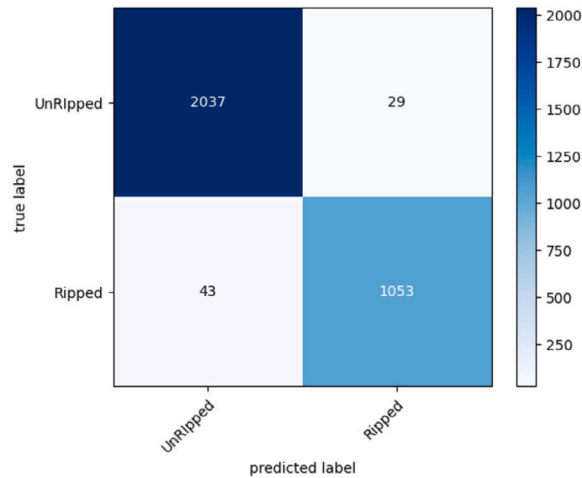
Video	Original Count	Predicted Count	Detection Rate (%)
Video 1	358	303	84.6
Video 2	589	441	74.8

Table 5
Results of per tree yield of Oranges.

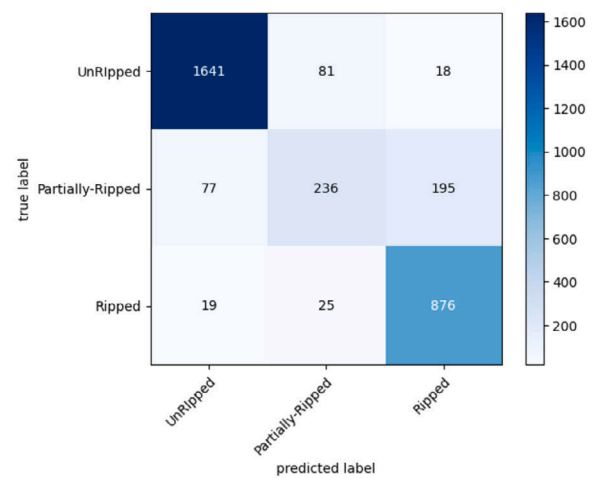
Video	Tree ID	Original Count	Predicted Count	Detection Rate (%)
Video 1	Tree 1	358	303	84.6
Video 2	Tree 1	196	120	61.2
Video 2	Tree 2	286	258	90.2
Video 2	Tree 3	107	63	58.8

few objects in mapping the objects when the object is moving fast in the video. However, this parameter can be manually adjusted in the algorithm by varying the value of maximum distance while calculating the distance between the objects.

Video 1 comprises of a single orange tree with sparse occlusion due to leaves and is under good lighting. Whereas the Video 2 consists of 3 trees in a row. The lighting conditions of this video are below satisfactory and trees have different densities of leaves on them. The results of orange counting on complete row and per tree yield are given in Table 4 and Table 5. The proposed solution can detect 84.6% oranges in Video 1 whereas the detection rate dropped to 74.8% in Video 2. Furthermore, for per-tree yield, the highest detection rate of 90.2% is obtained for tree 2 in video 2, and the lowest detection rate of 58.8% is recorded for tree 3 in video 2. Quantitative analysis of the algorithm shows a huge variance in the results. However, the manual inspection of these videos



(a) Confusion matrix of MobileVit on dataset consists of 2 classes



(b) Confusion matrix of MobileVit on dataset consists of 3 classes

Fig. 9. Confusion matrices of MobileVit.

for qualitative analysis shows that the algorithm performs best when there is low occlusion. Videos are shared [here](#).

6. Limitation and future work

The proposed system demonstrates effective performance when the camera moves in a linear direction. However, this approach limits visibility, particularly in capturing a comprehensive view of trees where some fruits remain hidden behind leaves. Consequently, the model fails to detect these occluded fruits due to their absence from the video feed, highlighting a significant drawback of the current solution. To enhance detection rates, an improved system could be devised that employs a 360-degree view to ensure all fruits are within the camera's field of view.

Another critical issue is the lack of proper handling for camera movements that backtrack and may inadvertently record fruits already captured. This oversight increases the likelihood of recounting fruits, necessitating the development of a robust methodology to address this repetition problem effectively. Such a methodology would ensure accurate fruit yield estimation by preventing redundant data collection.

An alternative enhancement involves leveraging airborne technology for fruit yield estimation. By capturing more stable video streams from a consistent height and speed, airborne systems offer greater efficiency and reliability compared to ground-based approaches. Integrating this technology not only simplifies deployment but also enhances the overall effectiveness of fruit yield estimation processes.

Furthermore, expanding the application of this system to mobile platforms would democratize farmers' access to fruit yield estimation. Deploying the solution via mobile applications makes it more accessible and cost-effective, empowering farmers to estimate yields conveniently using their smartphones. This accessibility not only enhances usability but also supports broader adoption and utilization in agricultural settings.

CRedit authorship contribution statement

Amna Hassan: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Data curation, Conceptualization. **Rafia Mumtaz:** Writing – original draft, Visualization, Validation, Supervision, Methodology, Investigation, Conceptualization. **Vasile Palade:** Writing – review & editing, Visualization, Validation, Methodology, Data curation, Conceptualization. **Arsalan Amin:** Writing – review & editing, Visualization, Validation, Supervision, Methodology, Investigation, Data curation. **Zahid Mahmood:** Writing – review & editing, Visualization, Validation, Resources, Investigation, Data curation.

Noorullah Khan: Writing – review & editing, Validation, Resources, Investigation, Data curation. **Muhammad Noman:** Writing – review & editing, Visualization, Validation, Resources, Investigation, Data curation. **Muhammad Imran:** Writing – review & editing, Visualization, Validation, Resources, Investigation, Data curation. **Santichai Wicha:** Writing – review & editing, Visualization, Validation, Methodology, Investigation, Conceptualization.

Ethical approval

The project does not involve any human or animal as a subject to be studied. Therefore, ethical approval is not applicable.

Consent to participate

The project does not require any involvement of living beings, therefore the consent to participate is not applicable.

Consent to publish

All authors of the paper have given the consent to publish this work.

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Declaration of competing interest

The authors declare that they have no conflict of interest.

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Availability of data and materials

The data can be made available by requesting the authors through email.

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